

CPSC 330 Lect

Recommen

Systems

Announcements

Show of Hands

What percentage of watch time on YouTube do you think comes from recommendations?

- A. 30%
- B. 40%
- C. 60%
- D. 70%
- E. 90%

Based on YouTube Product Chief at CES 2015, this number may have changed since then!

What is a recommendation system?

A recommendation system suggests products or content that a user is likely to consume.

Example: Recommendation systems

- A user visits Amazon to shop.
- Amazon knows:
 - what the user viewed or purchased
 - what similar users bought
- **The goal:** recommend items that maximize engagement or sales.
- There's no single "right" label. The goal is to predict user behaviour.

Why should we care?



- Recommends what to read, eat, or watch
- Central to the success of many companies like Netflix, YouTube, and Spotify.

Why recommendation

- They help users navigate information overload.
- Without them, finding the right item would require sifting through a large number of options.
- Recommendations reduce effort and improve user experience.



The other side: Filter bubbles

- Recommenders often amplify what you like or what similar users like.
- This can create filter bubbles, limiting exposure to diverse content.
- Probably harmless in shopping, but can have **serious consequences in domains like news and science, where diverse viewpoints are important**.

Data and prob setup

What data do we need

To build a recommender, we typically use

- User–item interactions (ratings, clicks, purchases)
- Item or user features (e.g., genre, price)
- Historical data (purchase or viewing history)

Problem formulation

- We have N users and M items.
- Observed data = interactions:
 - Movie ratings (Netflix)
 - Song plays (Spotify)
 - Product purchases (Amazon)

Goal: predict unobserved interactions.

The utility matrix

- Rows = users, columns = items
- Each entry y_{ij} = user i 's interaction (e.g. rating) with item j

				
		Item 1	Item 2	Item 3
	User 1	?	?	2
	User 2	3	?	?
	User 3	?	5	4
	User 4	?	?	?
	User 5	?	?	?
	User 6	?	5	4

Sparsity

- The utility matrix is mostly empty.
- Each user interacts with only a small number of items.
- Examples:
 - Netflix users rate only a few shows out of thousands.
 - Amazon shoppers review only a handful of products.

What do we predict?

We aim to fill in the missing entries. In predict ratings or preferences the user yet.

				
		Item 1	Item 2	Item 3
	User 1	?	?	2
	User 2	3	?	?
	User 3	?	5	4
	User 4	?	?	?
	User 5	?	?	?
	User 6	?	5	4

Rating prediction $\neq$ reg

Regression

`$$ \begin{bmatrix}`
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Rating

[illegible]

Main approaches

- Collaborative filtering
 - “Unsupervised” learning
 - We only have labels y_{ij} (rating of user i for item j).
 - We learn latent features.
- **Content-based recommenders** (today’s focus)
 - Supervised learning
 - Extract features x_i of users and/or items building a model to predict y_i given x_i .
 - Apply model to predict for new users/items.
- Hybrid
 - Combining collaborative filtering with content-based

Evaluating Recommender Systems

How do we evaluate recommendations?

- Is there a **single correct answer** for recommended or what rating should
 - **Not really!**
- Still, we need ways to **compare different** and measure their usefulness.

Why evaluation matter

- We'll experiment with different ways **entries** in the utility matrix.
- Even though recommendations are s need **quantitative metrics** to judge
 - How well do our predictions match behavior?
 - Which model performs better?

RMSE for rating prediction

- RMSE, which measures how close predicted ratings are to the actual ratings, is one of the commonly used metrics to evaluate recommendation algorithms.
- In **2006**, Netflix launched the **Netflix Prize** competition.
- They released a dataset of **100 million movie ratings** to the first team that improved Netflix's existing algorithm by **RMSE** on a held-out test set.

Netflix Prize				
COMPLETED				
Home Rules Leaderboard Update Download				
Leaderboard				
Showing Test Score. Click here to show quiz score				
Display top 20 leaders.				
Rank	Team Name	Best Test Score	% Improvement	Best Submit Time
Grand Prize - RMSE = 0.8567 - Winning Team: BellKor's Pragmatic Chaos				
1	BellKor's Pragmatic Chaos	0.8567	10.06	2009-07-26 18:18:28
2	The Ensemble	0.8567	10.06	2009-07-26 18:38:22
3	Grand Prize Team	0.8582	9.90	2009-07-10 21:24:40
4	Opera Solutions and Vandelay United	0.8588	9.84	2009-07-10 01:12:31
5	Vandelay Industries I	0.8591	9.81	2009-07-10 00:32:20
6	PragmaticTheory	0.8594	9.77	2009-06-24 12:06:56
7	BellKor in BigChaos	0.8601	9.70	2009-05-13 08:14:09
8	Dace	0.8612	9.59	2009-07-24 17:18:43

Clicker 16.1

Select all of the following statements which are **True**

- A. In the context of recommendation systems, the sh matrix and train utility matrix are the same.
- B. RMSE perfectly captures what we want to measure recommendation systems.
- C. It would be reasonable to impute missing values in taking the average of the ratings given to an item.
- D. In KNN type imputation, if a user has not rated any strategy would be recommending them the most p

Clicker 16.2

Select all of the following statements which are **True**

- A. In content-based filtering we leverage available item similarity between users.
- B. In content-based filtering you represent each user in terms of items.
- C. In the set up of content-based filtering we discussed, you would have problems predicting ratings for that user.
- D. In content-based filtering if a user has a number of ratings in the training utility matrix but does not have any ratings in the validation set, we won't be able to calculate RMSE for the validation set.

Baseline approaches

- Global average baseline
- Per-user average baseline
- Per-item average baseline
- Average of 2 and 3
 - Take an average of per-user and p
- *k*-Nearest Neighbours imputation

Content-based filtering

					
Eva	?	?	?	5.0	?
Jim	?	?	4.0	5.0	?
Pat	5.0	5.0	?	?	4.0
Sam	5.0	5.0	?	?	5.0

Eva’s profile

	Animation	Drama	Documentary	Rating
	0	0	1	5.0
	0	0	1	5.0
	1	1	0	1.0

Pat’s profile

	Animation
	1
	1
	0

...

...

iClicker

Select all of the following statements w

- A. In content-based filtering we leverage content features in addition to similarity between items.
- B. In content-based filtering you represent items in terms of **known** features of items.
- C. In the set up of content-based filtering discussed, if you have a new movie, you face problems predicting ratings for that movie.
- D. In content-based filtering if a user has ratings in the training utility matrix but doesn't have any ratings in the validation utility matrix, we won't be able to calculate RMSE on the validation utility matrix.

Ethics of Recommender Systems

We should be very mindful of the drawbacks on recommender systems.

What are some ethical considerations for recommender systems?

What comprises a recommender system?

What does the problem formulation look like?
What tools would/should we use to create a recommender system?

Group Work: Class Demo Coding

For this demo, each student should [click](#) follow along with the demo from today.
All credit to Dr. Varada Kolhatkar for putting this together!